

Driving as Side Task

Exploring Intuitive Input Modalities for Multitasking in Automated Vehicles

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ABSTRACT

Through vehicle automation, the human is more and more able to execute non-driving-related activities (NDRAs). However, control interfaces are still needed to allow human intervention and foster a sense of control. This paper presents a driving simulator experiment ($N = 20$) with five likely NDRAs (being idle, eating, smartphone use, conversation, listen to music) where participants freely intervened in the driving process, using either touch, voice, or mid-air gesture interaction. We found that NDRAs determine input modality choices. Generally, participants tended to avoid the modalities demanded by the NDRAs, e.g., voice interaction becomes less frequent while listening to music. Contrary, touch interaction increased during smartphone use, indicating that users tend to stick to a known interaction style during high multitasking workload. Overall, we recommend designing future vehicle interiors and interfaces that rely on multimodal interfaces to account for the diversity of situations and activities while driving automated.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; **User studies**; **Interaction techniques**.

KEYWORDS

Automated Vehicles, Non-Driving-Related Activities, Driving-Related Activities, Maneuver-based Interventions, Control, Multitasking, Multimodal Interaction, Touch, Voice, Mid-Air Gestures.

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1 INTRODUCTION

Automated cars will change the role of the driver. Traditionally the driver is responsible for the driving task. With increasing technological progression and the integration of driving assistant systems into cars, the driving task shifts more and more to the system.

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Reaching certain degrees of automation (cf., SAE level 3–5 [22]), the system can complete the driving task itself, at least for a certain time, depending on the level. From then on, the human driver becomes the passenger (e.g., from entering until exiting the highway) and is free to engage in non-driving-related activities (abbr.: NDRAs): one can do work, entertainment, or social activities during travel. Nevertheless, even in autonomous driving phases, human-vehicle interaction is still necessary to provide mechanisms to the human driver to intervene in the driving task, e.g., to adapt driving parameters, take cooperative maneuvers, or foster a sense of control. The NDRAs highly influence the context of such driving-related activities (abbr.: DRAs). When the occupants want to intervene in the driving process, they must shift their attention to the new task, and the current activity must be interrupted. Interference between NDRA and DRA because of similar requirements for spatial/verbal cognitive resources (cf. [25]), might lead to performance breakdowns. Thus, it is crucial to understand the interplay between NDRA and DRAs in automated vehicles. While the definition [19], prediction [5, 12, 13, 18], and automated recognition [17] of NDRAs, as well as their influence on take-over request performance [6, 10, 15, 20, 27] have been studied recently, their influence in less critical multitasking scenarios is still unexplored. This paper therefore explores the influence of highly anticipated non-driving-related activities (main task) on the driver/passenger's modality choices when intervening in the driving process (side task) in a driving simulator experiment ($N = 20$). To interact in a subjectively natural and comfortable way, participants could freely choose between touch, voice, and mid-air gesture modalities. We were interested in resulting patterns of interaction and possible differences in multitasking performance. Thereby, we hypothesize that (1) NDRAs would influence the chosen input modalities for control interventions, and (2) the selected modality would be different from the modality used for the main task. Our study helps to understand better which interaction modalities are *used naturally* in the context of different NDRAs. Further, it is a first step towards the *activity-optimized development* of in-car HMIs for driving-related control interventions in future automated cars.

2 RELATED WORK

In the following, we distinguish DRAs from NDRAs and show how control in automated cars will change. The growing automation will lead to new activities, and new activities will change the user requirements for in-car interfaces.

2.1 Non-Driving-Related Activities

In conventional cars, one can divide the driving task into three categories (cf. Bubbl [1]): 1) transportation-related tasks which are

required to reach a destination (e.g., use the steering wheel), 2) transportation-related tasks which are optional to reach a particular destination (e.g., use the windscreen wiper), and 3) non-transportation-related tasks (e.g., use media control).

With the system's growing responsibility for the primary and secondary driving tasks through automation, the former driver becomes more and more a passenger in automated cars, and tertiary tasks become more important than before. To simplify the categorization of tasks in automated vehicles, we use the terms *Driving-Related Activities or Tasks* and *Non-Driving-Related Activities or Tasks* as proposed by Pfleging & Schmidt [19]. Expected NDRA are, for example [5, 12, 13, 18]: Watching out of the window, talking to other passengers, using the smartphone, eating, listen to music/radio, taking a nap, knitting, or office work. These activities require different physical and cognitive resources and influence the human vehicle user's capabilities and preferences to control the car.

2.2 Control During Autonomous Driving

An essential requirement of cooperative driving is directability [24]. From a functional perspective, use cases where the human might want to intervene in an automated ride are, for example [23]: Influencing the route taken, adjusting the driving style of the vehicle (e.g., to accelerate more quickly at the traffic light), or overtaking of slow traffic before the highway exit. From a psychological perspective, it is beneficial to provide a control interface possibility. Compared to being passive in autonomous driving, the possibility to exercise control increases driving pleasure and attractiveness [9] and the acceptance of the system [16].

Traditional controls (steering wheel and pedals) might not be the best option because they require detailed awareness of the driving situation on the operational level, e.g., the exact steering angle for a DRA. A more comfortable way of controlling the car could be maneuver-based or trajectory-based control [7] because operational aspects of the driving task remain with the vehicle. We refer to these concepts as *Trajectory-based Interventions* and *Maneuver-based Interventions* (abbr.: MBI). Such cooperative concepts combine the advantage of automated driving with user needs and requirements. Because these driving interventions are intentional actions of the driver/passenger, interfaces for maneuver control commonly rely on explicit and fast interaction modalities, e.g., touch [8, 14], voice, or mid-air gestures [3, 4]. However, objective utility and subjective perception of these interfaces are heavily dependent on the interaction context and, thus, on the NDRA.

2.3 Multitasking and Multimodal Interaction

During autonomous travel times, vehicle occupants are likely to perform NDRA. When they are occupied with a particular NDRA, they spend physical and mental resources for that task, e.g., reading a book requires the physical effort to hold the book and the mental effort to decode and understand the written letters – eyes and hands, as well as attention, are dedicated to this task. Control interventions potentially use the same and thus might interfere with these resources. Wickens [26] defines four dimensions of resource sharing in multitasking settings: 1) processing dimension

(perception, cognition, response), 2) codes of processing dimension (spatial vs verbal), 3) modalities dimension (auditory vs visual perception), and 4) visual channels (focal vs ambient). The less a multitasking procedure occupies the same dimensions, the better the later performance.

Roider et al. [21] found for input modality switches for side tasks during driving, that transitions between interaction modalities can increase efficiency compared to unimodal conditions and that switching modalities does not increase overall task completion time. Further, their results indicate that distractions from the main task are determined by the used modality of the side task, not by transition processes. Once distraction from the main task (driving) is no longer the primary concern of interaction, the system should promote modality switches between tasks so that users could benefit from individual modality's efficiency and suitability for a particular task. Thus, it might be useful to switch control intervention modality during an NDRA, e.g., voice control would not be the best choice while listening to music, and touch not the best choice while using the smartphone.

3 EXPERIMENT

We conducted a within-subject driving simulator experiment to explore modality choices for control interventions in multitasking scenarios. On a short test track, participants performed five possible non-driving-related activities (counterbalanced order) during which they intervened four times in the driving process. Each time they could choose between either voice, touch, or mid-air gestures (resulting in 20 choices per participant or 80 choices per condition).

Sample. We conducted the study at the University of Applied Sciences Ruhr West. Twenty computer science students ($m = 17$; $f = 3$) with an average age of 22.45 years ($SD = 4.55$; $Min = 18$; $Max = 33$) participated.

Task A: Driving-Related Activity. We told participants that they would ride a safe driving car (cf. driving phases during SAE level 3–5) operating in a cooperative driving mode that requires their control. We instructed participants to adhere to the following routine during the rides, intervening in the autonomous driving process: A1) start the vehicle (*start maneuver*), A2) command the car to drive onto the motorway from the rest area (*lane change maneuver*), A3) control the vehicle to leave the highway at the next rest area (*lane change maneuver*), and A4) park at a rest area (*park maneuver*). Participants were free to choose an interaction modality out of voice, touch, and mid-air gestures. We selected these modalities because they allow for direct interaction. For touch interaction, we used a maneuver-board interface on a touchpad (identical to [14]), and for voice and mid-air gesture interfaces, we used a user-defined command set (identical to [3]).

Task B: Non-Driving-Related Activity. Based on studies on expected NDRA (cf. [5, 12, 18]), we discussed frequent activities concerning their potential impact on driving-related interventions. Afterward, we selected five prototypical activities for our experiment that would involve participants in different ways. For the selected tasks, we gave participants the following instructions in addition to Task A (cf. Table 1).

Table 1: Instructions given for each situation and situational demands for multitasking

NDRA	Additional Instructions	Resource Involvement
Do Nothing	-	-
Eating	"Please choose one of the following snacks [saltines, crackers, flips] to eat while driving. Please take the entire bowl and don't put it down anywhere."	Manual demand, and limited ability to speak
Smartphone	"Please pick up your cell phone and go to YouTube to watch a cat video while driving." End of Smartphone situation (if video too long): "Okay, that's it! Please stop the video and put your smartphone aside."	Manual and visual demand, high cognitive involvement
Conversation	"And now I'll distract you a bit with a conversation..." "What are you studying here with us? ...why?", "Which was your favorite module so far? ...why?"	Auditory demand, cognitive involvement
Music	"Now please start with the echo dot music of your choice."	Constant auditory demand

**Figure 1: Setup - The participant sits in driving simulator main seat, the main investigator acts as a co-passenger, and the wizard simulates the system in the back.**

Setup and Wizard of Oz. We used a driving simulator (Software: SILAB 6.0) with sounds for the study (see Figure 1). Sounds, e.g., engine sounds, were on a quiet level so that conversation and listen to music were possible without interference. The steering wheel was mounted with a touch tablet (Microsoft Surface Pro) to select driving maneuvers. We used the Wizard of Oz (abbr.: WoZ) approach [2] to simulate the automated system behavior. The touch input and speech and gesture input were interpreted by a wizard that sat in the back of the scene. We told participants that they would interact with a real system prototype. After the test, no participant reported that (s)he noticed the mediation. The leading investigator (not the same person as the wizard) instructed the participants and acted as the co-passenger in the conditions *Eating* (reaching food) and *Conversation* (asking questions).

Procedure and Measurements. First, the leading investigator briefed participants and asked for their informed consent. The study then started with Task A (intervention) tutorial and participants got a quick introduction to the driving simulator, and we explained the necessary commands. Further, they drove a tutorial track to overtake trucks (lane-change maneuvers) to get comfortable with the different control modalities. Afterwards, each participant experienced five experimental conditions (same route) in a counterbalanced within-subject design. In each condition, the leading investigator began with standardized instructions to introduce Task B. Then, he left the scene or acted in the conditions "Eating" and "Talking" as a co-passenger. During the non-driving-related activities (Task B), participants intervened four times (see Task A) in the autonomously driving cars' behavior (operated by the

system wizard). Participants had the free choice to execute these maneuvers either via touch or speech or mid-air gesture control. A third investigator who solely filled out the study protocol recorded all modality choices. Completing a condition took about 2 minutes. After each run, participants filled out the NASA-TLX ([11]; 6 items on a 100-point slider scale) questionnaire to measure their subjective task load in the situations. After five runs, they filled out a final questionnaire regarding general feedback and preferences for a particular modality. Then, participants got debriefed. Overall, the study took about 30 minutes.

4 RESULTS

For each NDRA, we analyzed and compared the observed interaction choice pattern in terms of modality shares and consistency. Further, we looked into the subjectively reported interaction workload and the overall preference for an input style.

4.1 Modalities' Shares

Table 2 shows the observed interaction frequencies. To determine the influence of a non-driving-related activity on the modality choice for a driving-related intervention, we conduct Pearson's chi-squared tests to compare the goodness of fit of *Eating*, *Smartphone*, *Conversation*, and *Music* conditions' distributions with the *Do Nothing* condition's distribution that serves as a baseline. The distribution of chosen interaction modalities in the *Eating* ($\chi^2(2, N = 80) = 2.51, p = .29$) and *Conversation* ($\chi^2(2, N = 80) = 2.82, p = .25$) conditions are not different from the baseline, while *Smartphone* ($\chi^2(2, N = 80) = 6.70, p < .05$) and *Music* ($\chi^2(2, N = 80) = 6.13, p < .05$) have

Table 2: Observed modality choices – distribution of modalities’ (voice, touch, mid-air gestures) use frequency in each experimental condition (activities), comparison of distributions between *Do Nothing* and other activities, significant deviations from the baseline with $p < .05$ are marked through *-signs; variability between modalities indicated through Flexibility (cf. equation 1) and Change Ratio (cf. equation 2)

Activity	Interaction Frequency n (%)				Interaction Variability <i>mean, sd</i>	
	Voice	Touch	Mid-Air Gesture	Σ	Flexibility	Change Ratio
Do Nothing	19 (24)	35 (44)	26 (32)	80 (100)	.25 ± .3	.23 ± .31
Eating	13 (16)	39 (49)	28 (35)	80 (100)	.25 ± .30	.22 ± .27
Smartphone*	19 (24)	45* (56)	16* (20)	80 (100)	.28 ± .30	.25 ± .30
Conversation	22 (27)	39 (49)	19 (24)	80 (100)	.18 ± .34	.15 ± .31
Music*	10* (12)	43 (54)	27 (34)	80 (100)	.23 ± .34	.23 ± .36
Overall	83 (21)	201 (50)	116 (29)	400 (100)	.24 ± .32	.22 ± .31

significantly different distributions. Inspecting the values in the *Smartphone* and *Music* conditions, we see comparable frequencies for voice, lower frequencies for mid-air gestures, and higher frequencies for touch interaction. Bonferroni-corrected Binominaltests ($p < .025$) for these conditions show that participants used significantly less voice interaction in the *Music* condition and significantly more touch and less mid-air gesture interaction in the *Smartphone* condition. Overall, touch control (50%) was used nearly as twice as often as voice (21%) or mid-air gesture control (29%).

4.2 Modalities’ Consistency

Regarding the consistency of participants’ modality choices, we looked at two indicators that we define in the following a) the *Flexibility*, and b) the *Change Ratio* per condition. The interaction Flexibility and Change Ratio provide robust and easy to interpret indicators for variation in modality choice over time, in a set I of n subsequent interactions (in our case: $n = 4$). The Flexibility indicator shows the use of distinct modalities compared to all interaction types in the conditions, shows if participants’ stick to one interaction type over time ($\text{Flexibility}(I) = 0$) or make use of the whole range of possible interaction types ($\text{Flexibility}(I) = 1$). We define the Flexibility for a set of interactions I where each interaction has M possible modalities:

$$\frac{|I \cap M|}{M-1} - \frac{1}{M-1} \quad (1)$$

We define the Change Ratio for a set I consisting of n subsequent interactions as the number of times the modality m of an interaction is unequal to the previous interaction modality:

$$\sum_{i=1}^{n-1} \frac{m_{I_i} \neq m_{I_{i+1}}}{n-1} \quad (2)$$

The Change Ratio shows the general variability of interactions over time, independent from particular modalities. An example: If participant A uses an interaction sequence of $\{\text{Voice} \rightarrow \text{Touch} \rightarrow \text{Voice} \rightarrow \text{Touch}\}$ and participant B $\{\text{Voice} \rightarrow \text{Touch} \rightarrow \text{Touch} \rightarrow \text{Touch}\}$, both have the same Flexibility (A: 0.5, B: 0.5), but the Change Ratio (A: 1, B: 0.33) of A is higher. Table 2 shows the variability in our sample: About every fifth time ($M_{\text{Flexibility}} = .22, SD =$

.32), participants chose another follow-up modality and mostly stick to one or two modalities ($M_{\text{ChangeRatio}} = .24, SD = .31$).

4.3 Intervention Task Workload during NDRAs

Figure 2 shows the mean NASA-TLX responses. We used a repeated-measures ANOVA (sphericity checked with Mauchly’s test, $p > .05$; and degrees of freedom corrected after Greenhouse-Geisser, if $\epsilon \leq .75$, otherwise after Huynh-Feldt) to determine the effects of the situation on task performance. We conducted Bonferroni-corrected posthoc tests to detect differences between conditions (see Figure 2). Looking into the NASA-TLX subscales, we found no effects for Physical Demand ($F(1.69, 32.18) = 2.54, p = .1, \eta^2 = .12$), Temporal Demand ($F(2.28, 43.23) = 1.26, p = .3, \eta^2 = .06$), and Frustration ($F(4, 76) = 1.58, p = .19, \eta^2 = .08$). For Mental Demand ($F(4, 76) = 5.99, p < .001, \eta^2 = .24$), Performance ($F(4, 76) = 3.61, p < .05, \eta^2 = .16$), and Effort ($F(2.67, 50.71) = 4.94, p < .01, \eta^2 = .21$) we found significant effects with a strong effect size after Cohen ($f_{\text{MentalDemand}} = 0.56; f_{\text{Performance}} = 0.44; f_{\text{Effort}} = 0.52$). We found a significant main effect of the situation on the mean Raw-TLX load ($F(4, 76) = 4.55, p < .01, \eta^2 = .19$). After Cohen, the effect is strong ($f = 0.49$). Posthoc tests reveal that the *Smartphone* condition leads to higher perceived overall load than the *Do Nothing* condition. We found no significant correlation between the modality choices and overall task load.

4.4 Overall Preference for a Modality

Participants subjective rating for voice ($M = 4.30, SD = 0.73$), touch ($M = 4.00, SD = 1.26$), and mid-air gesture ($M = 3.70, SD = 1.22$) input modalities after the experiment is high. A repeated-measures ANOVA (sphericity checked with Mauchly’s test, $p > .05$) reveals no significant main effect of the modality on rating ($F(2, 38) = 1.81, p = .18, \eta^2 = .09$). Hence, there is no clear preference for a particular modality yet visual inspection shows a non-significant trend with voice being higher rated than touch and touch higher than mid-air gestures.

5 DISCUSSION

In the following, we discuss our findings and point out limitations and directions for future work on multitasking during autonomous driving.

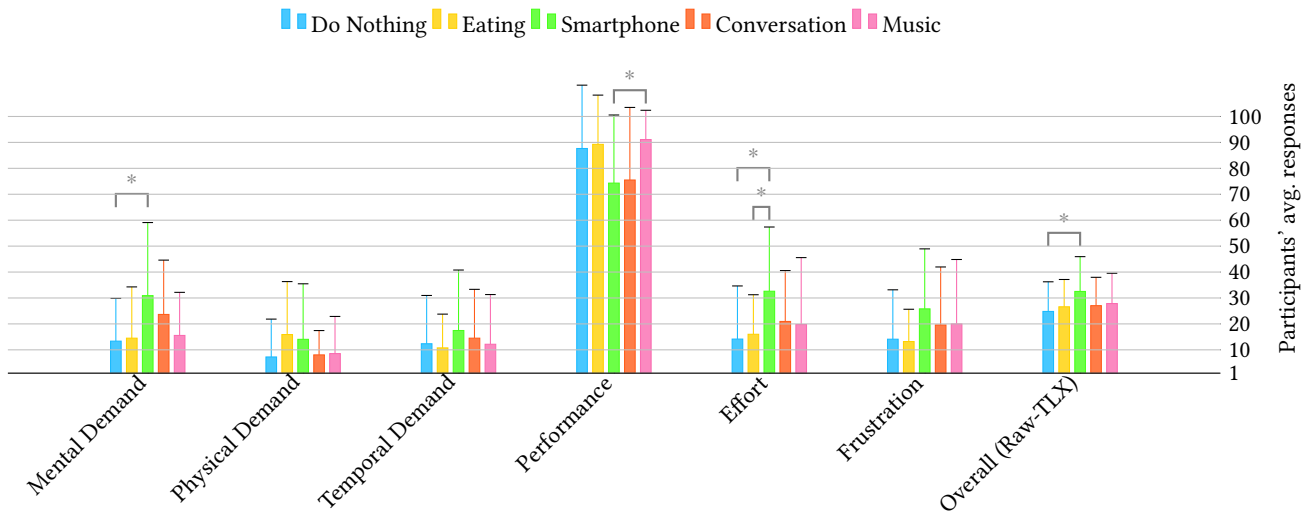


Figure 2: Multitasking: Results from the NASA-TLX Questionnaire, error bars indicating standard deviation, significant differences with $p < .05$ are marked through *-signs

In this experiment, we did not control the modality switches but gave participants a choice. This kind of procedure allows for an economic sample size and an analysis of switching preferences/patterns. It might be useful for exploring multitasking with multimodal interfaces in other contexts, too. However, a more extensive follow-up study could investigate modality switches in fixed pairs. We hypothesized that (1) NDRA would influence the chosen input modalities for control interventions, and (2) the selected modality would be different from the modality used for the main task. We could fully confirm the first and partly confirm the second assumption with our experiment.

In terms of multitasking modality switch patterns, we saw that participants use more than one interaction modality per condition (change ratio $> .22$ / flexibility $> .24$). These modality changes could be explained by the findings of Roider et al. [21] that transitions between interaction modalities can increase efficiency compared to unimodal conditions. Further, we observed that in terms of overall frequency, *Eating* (chewing) and *Music* (acoustic interference) conditions lead to comparably less voice interaction, while for the *Smartphone* (manual demand) and *Conversation* (supporting gestures) conditions, mid-air gesture input is less used. These findings indicate that users tend to avoid resource conflicts through an NDRA occupied modality as input for the control intervention, which is in line with the second hypothesis. Thus, future vehicles should provide redundant control mechanisms to the users because they will likely perform various NDRA during a trip. Through tracking the occupants' activities, the system could promote the most feasible modality choice. Future work could also investigate the specific promotion of side-task interaction (control interventions) for common NDRA and develop strategies to train users to make sure they use each interaction modality's full potential from the beginning on, e.g., through User Onboarding.

We observed a high increase of touch interaction during the smartphone condition, which is, in contrast, not in line with the second assumption. In terms of overall multitasking load, we

found the smartphone condition to be different from the baseline. This could be explained by the multiple resources theory because the smartphone condition had a high combined spatial (typing/watching) and verbal (typing/listening) demand on all processing stages. Given that smartphone use is the most anticipated NDRA in automated vehicles, it will be challenging to design systems in a way that shifts the users' attention and facilitate the switch to other tasks. Future concepts thus might integrate the use of the smartphone in multitasking procedures (cf. Wintersberger et al. [27]). Given the higher total multitasking workload in the smartphone condition, we assume that users have less free cognitive resources to use non-familiar interaction styles and, therefore, instead stick to known patterns. Touch interaction, known from everyday smartphone use, might feel more comfortable. This bias would also explain the general tendency of using touch interaction, despite the mid-ranking in the users' overall modality preferences. Future work could pick up this challenge look into the task switching efficiency, as an already demanding modality might decrease interaction efficiency (cf. Roider et al. [21]).

Beyond the previously discussed resource conflicts, other factors like social norms might influence the interaction behavior, e.g., eating while talking to somebody or something might be considered rude behavior, or using an "expressive" interaction style like mid-air gesture or voice interaction might be socially inhibited.

5.1 Limitations of Study

In our study, we used a WoZ system. The error rate of voice and mid-air gesture recognition systems will be higher for a real implementation, and findings might differ. Our sample was relatively small and homogeneous in this initial study, consisting mainly of young male students with a strong technical background. Future work should investigate a larger and more heterogeneous user sample to strengthen the generalizability of the findings.

6 CONCLUSION

During autonomous driving, vehicle occupants will perform NDRAs, and, sometimes, intervene in the driving process. The former relation of main and side tasks will be reversed. Driving will become an optional side task. Our study shows that NDRAs lead to different interaction modality choices while multitasking in the car, highlighting the importance of multimodal interaction design. For an intervention, users tend to avoid resource conflicts with an occupied modality, e.g., voice interaction becomes less frequent while eating or listen to music, and mid-air gestures become less frequent during conversations or smartphone use. In contrast, when the multitasking workload is comparably high, users tend to stick to an interaction style, as we found for touch interaction during smartphone use. Overall, we recommend implementing control concepts that rely on multimodal interfaces to account for future autonomous vehicles' diverse situations and activities. Despite the potential for more research on multitasking support in the vehicle, be it for car or entertainment control, our findings help sensitize the designers to the *NDR context of interaction* when planning new vehicle interiors and interfaces.

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